

A Quantitative Synthesis of Mercury in Commercial Seafood and Implications for Exposure in the United States

Background: Mercury (Hg) is a toxic metal that presents public health risks through fish consumption. A major source of uncertainty in evaluating harmful exposure is inadequate knowledge of Hg concentrations in commercially important seafood. Objectives: We examined patterns, variability, and knowledge gaps of Hg in common commercial seafood items in the United States and compared seafood Hg concentrations from our database to those used for exposure estimates and consumption advice. Methods: We developed a database of Hg concentrations in fish and shellfish common to the U.S. market by aggregating available data from government monitoring programs and the scientific literature. We calculated a grand mean for individual seafood items, based on reported means from individual studies, weighted by sample size. We also compared database results to those of federal programs and human health criteria [U.S. Food and Drug Administration Hg Monitoring Program (FDA-MP), U.S. Environmental Protection Agency (EPA)]. Results: Mean Hg concentrations for each seafood item were highly variable among studies, spanning 0.3–2.4 orders of magnitude. Farmed fish generally had lower grand mean Hg concentrations than their wild counterparts, with wild seafood having 2- to 12-fold higher concentrations, depending on the seafood item. However, farmed fish are relatively understudied, as are specific seafood items and seafood imports from Asia and South America. Finally, we found large discrepancies between mean Hg concentrations estimated from our database and FDA-MP estimates for most seafood items examined. Conclusions: The high variability in Hg in common seafood items has considerable ramifications for public health and the formulation of consumption guidelines. Exposure and risk analyses derived from smaller data sets do not reflect our collective, available information on seafood Hg concentrations.